

Name: _____

R. Hammack

Score: _____

Directions: Please answer all questions in the space provided. Use of calculators or any form of electronic communication device is strictly forbidden on this quiz.

1. Find a basis for the vector space of all 3×3 symmetric matrices. Explain your reasoning.

2. Consider the following subspace of $\mathbb{R}^4$: $W = \{(2s - 3t, s, t + s, t) : s \text{ and } t \text{ are real numbers}\}$
Find a basis for W . State the dimension of W .